

Höhere Technische Lehranstalt Shkoder  
Abteilung Systemtechnik – Schwerpunkt Data Science

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# VHGRAD

Entwicklung eines didaktischen Deep-Learning-Frameworks  
mit eigener Autograd-Engine

Diplomarbeit

Verfasst von

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# Technical Progression of the vhgrad Framework

- 1. Foundations of Neural Networks** Introduction to neural network concepts, historical context, and core components such as neurons, layers, and parameters.
- 2. Initial Implementations** Manual implementation of single neurons and multi-neuron layers with emphasis on mathematical foundations.
- 3. Vectorized Computation** Transition to NumPy-based implementations using vectors, matrices, batching, and efficient matrix operations.
- 4. Multi-Layer Network Construction** Encapsulation of dense layers into reusable classes and structured handling of training data.
- 5. Activation Functions** Implementation of linear and non-linear activation functions, including Sigmoid, ReLU, and Softmax.
- 6. Loss Functions and Evaluation** Integration of categorical cross-entropy loss and accuracy metrics for model evaluation.
- 7. Optimization and Backpropagation** Gradient-based optimization, derivatives, chain rule, and full backpropagation pipeline.
- 8. Optimizers** Implementation of SGD, learning rate decay, momentum, AdaGrad, RMSProp, and Adam.
- 9. Regularization Techniques** L1 and L2 regularization, dropout, and validation strategies to improve generalization.
- 10. Classification and Regression** Support for binary classification and regression tasks with appropriate loss functions.
- 11. Model Abstraction** Introduction of a unified model object encapsulating layers, losses, and optimizers.
- 12. Real-World Data Integration** Data loading, preprocessing, shuffling, batching, and training on real datasets.
- 13. Model Persistence and Inference** Saving and loading model parameters and enabling prediction on unseen data.